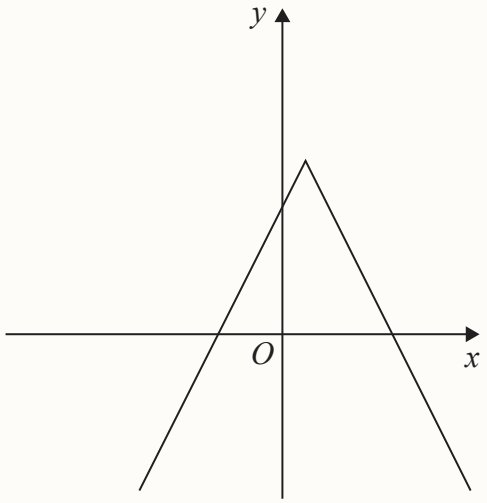



**Figure 2**

**In this question you must show all stages of your working.**

**Solutions relying on calculator technology are not acceptable.**

The graph shown in Figure 2 has equation

$$y = a - |2x - b|$$

where $a$ and $b$ are positive constants, $a > b$

(a) Find, giving your answer in terms of $a$ and $b$,

(i) the coordinates of the maximum point of the graph,

(ii) the coordinates of the point of intersection of the graph with the $y$-axis,

(iii) the coordinates of the points of intersection of the graph with the $x$-axis.

**(5)**

On page 24 there is a copy of Figure 2 called Diagram 1.

(b) On Diagram 1, sketch the graph with equation

$$y = |x| - 1$$

**(2)**

Given that the graphs $y = |x| - 1$ and $y = a - |2x - b|$ intersect at $x = -3$ and $x = 5$

(c) find the value of $a$ and the value of $b$

**(4)**